

```

> restart;
> # Exercise 1
> # a) find general solution
> ec:=diff(x(t),t)+x(t)=2/sqrt(Pi) * exp(-t^2-t);

$$ec := \frac{d}{dt} x(t) + x(t) = \frac{2 e^{-t^2-t}}{\sqrt{\pi}} \quad (1)$$

> dsolve(ec,x(t));

$$x(t) = (\operatorname{erf}(t) + c_1) e^{-t} \quad (2)$$

> # b) compute the primitives
> int(exp(t^2),t);

$$\frac{\sqrt{\pi} \operatorname{erfi}(t)}{2} \quad (3)$$

> int(2/sqrt(Pi)*exp(-t^2),t);

$$\operatorname{erf}(t) \quad (4)$$

> ?erf
> my_erf:=x->(2/sqrt(Pi))*int(exp(-t^2),t=0..x);

$$my\_erf := x \mapsto \frac{2 \cdot \left( \int_0^x e^{-t^2} dt \right)}{\sqrt{\pi}} \quad (5)$$

> evalf(my_erf(1));
0.8427007929 (6)
> evalf(erf(1));
0.8427007929 (7)
> restart;
> # Exercise 2
> # a) any solution of ec verifies the limit ?
> ec:=diff(x(t),t$2)+3*diff(x(t),t)+x(t)=1;

$$ec := \frac{d^2}{dt^2} x(t) + 3 \frac{d}{dt} x(t) + x(t) = 1 \quad (8)$$

> sol:=dsolve(ec,x(t));

$$sol := x(t) = e^{\frac{(\sqrt{5}-3)t}{2}} c_2 + e^{-\frac{(3+\sqrt{5})t}{2}} c_1 + 1 \quad (9)$$

> limit(rhs(sol),t=infinity);
1 (10)
> # b) the solution of the ivp verifies given condition?
> restart;;
> restart;
> ec:=diff(x(t),t$2)+4*x(t)=1;

```

$$ec := \frac{d^2}{dt^2} x(t) + 4 x(t) = 1 \quad (11)$$

```
> cond:=x(0)=5/4,D(x)(0)=0;
```

$$cond := x(0) = \frac{5}{4}, D(x)(0) = 0 \quad (12)$$

```
> sol:=dsolve({ec,cond},x(t));
```

$$sol := x(t) = \frac{1}{4} + \cos(2 t) \quad (13)$$

```
> x_func:=unapply(rhs(sol),t);
```

$$x_func := t \mapsto \frac{1}{4} + \cos(2 \cdot t) \quad (14)$$

```
> x_func(Pi);
```

$$\frac{5}{4} \quad (15)$$

```
> # c) check if equation admits polynomial solution
```

```
> restart;;
```

```
> restart;
```

```
> ec:=diff(x(t),t)=3*x(t)+t^3;
```

$$ec := \frac{d}{dt} x(t) = 3 x(t) + t^3 \quad (16)$$

```
> sol:=dsolve(ec,x(t));
```

$$sol := x(t) = -\frac{t^2}{3} - \frac{t^3}{3} - \frac{2t}{9} - \frac{2}{27} + e^{3t} c_1 \quad (17)$$

```
> RESTART;
```

RESTART (18)

```
> restart;
```

```
> salutare
```

salutare (19)

```
> restart;
```

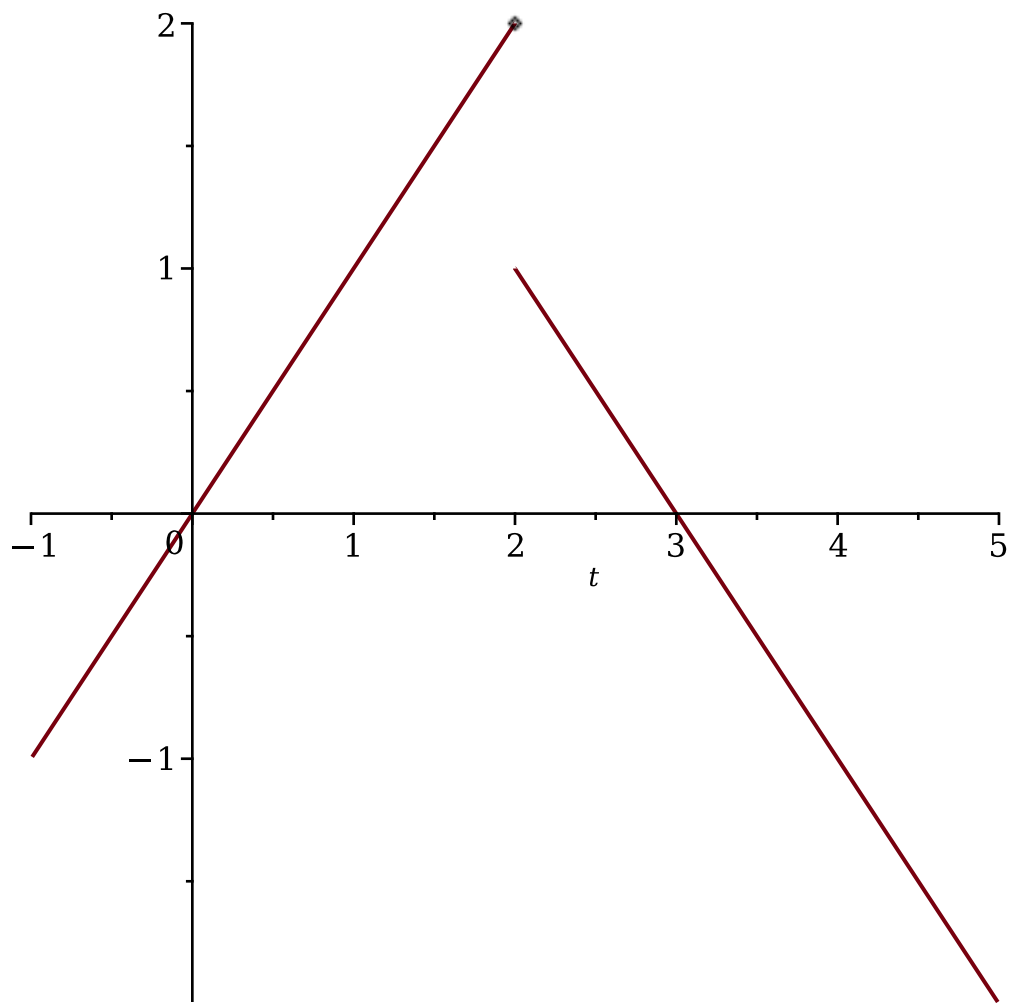
```
> # Exercise 3 - represent the graph of the function f in an
interval that contains the point where the function changes its
expression
```

```
> # a)
```

```
> f:=t->piecewise(t<=2,t,t>2,3-t);
```

$$f := t \mapsto \begin{cases} t & t \leq 2 \\ 3 - t & 2 < t \end{cases} \quad (20)$$

```
> plot(f(t),t=-1..5,discont=true);
```



```
> discontinuity(f(t),t);
```

$\{2\}$

(21)

```
> diff(f(t),t);
```

$$\begin{cases} 1 & t < 2 \\ \text{undefined} & t = 2 \\ -1 & 2 < t \end{cases}$$

(22)

```
> #b)
```

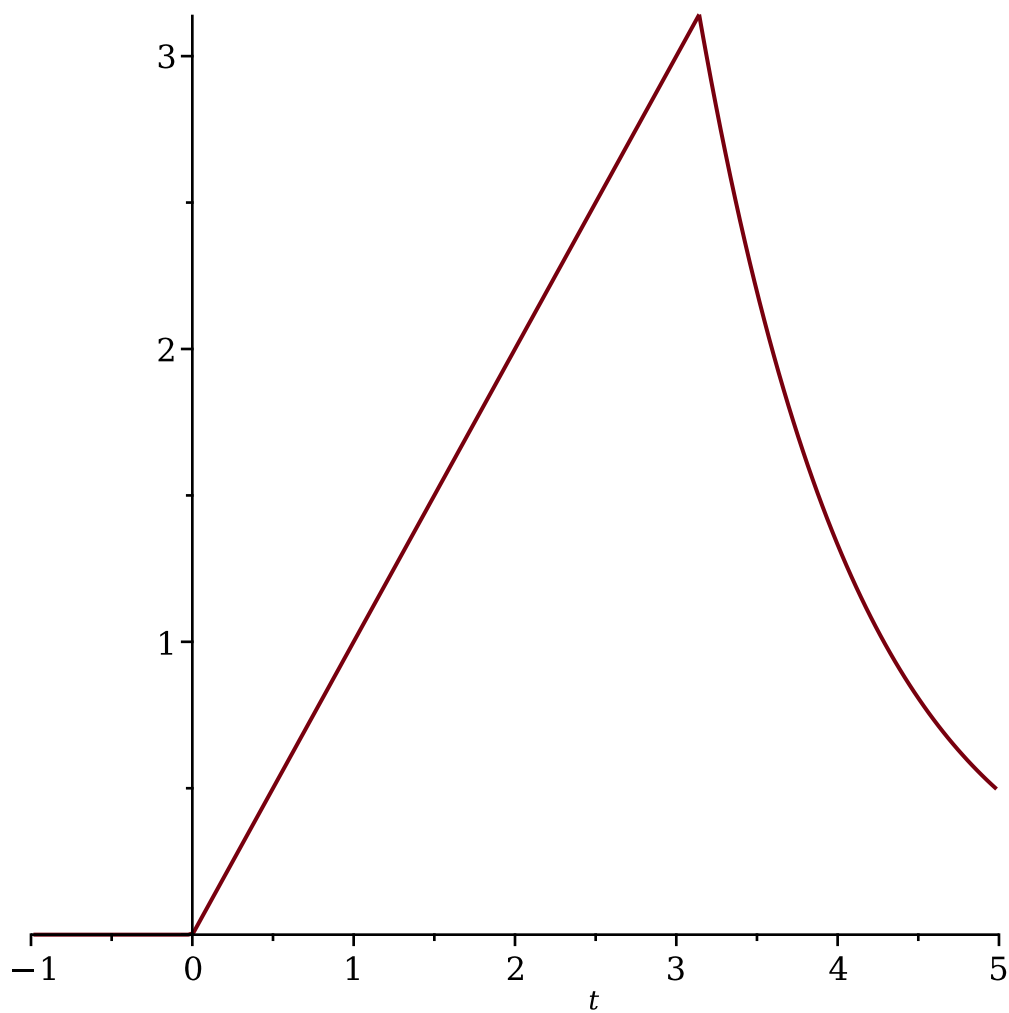
```
> restart;
```

```
> f:=t->piecewise(0<=t and t<=Pi,t,t>Pi,Pi*exp(Pi-t));
```

$$f := t \mapsto \begin{cases} t & 0 \leq t \leq \pi \\ \pi \cdot e^{\pi-t} & \pi < t \end{cases}$$

(23)

```
> plot(f(t),t=-1..5);
```



```
> discontinuity(f(t),t);
```

$\{0, \pi\}$

(24)

```
> diff(f(t),t);
```

$$\begin{cases} 0 & t < 0 \\ \text{undefined} & t = 0 \\ 1 & 0 < t < \pi \\ \text{undefined} & t = \pi \\ -\pi e^{\pi-t} & \pi < t \end{cases}$$

(25)

```
> # Exercise 4
```

```
> restart;
```

```
> # a) find solution of the IVP, plot the graph on the interval (0, Pi)
```

```
> ec:=diff(x(t),t$2)+x(t)=t;
```

$$ec := \frac{d^2}{dt^2} x(t) + x(t) = t$$

(26)

```
> cond:=x(0)=0, D(x)(0)=1
```

$$cond := x(0) = 0, D(x)(0) = 1 \quad (27)$$

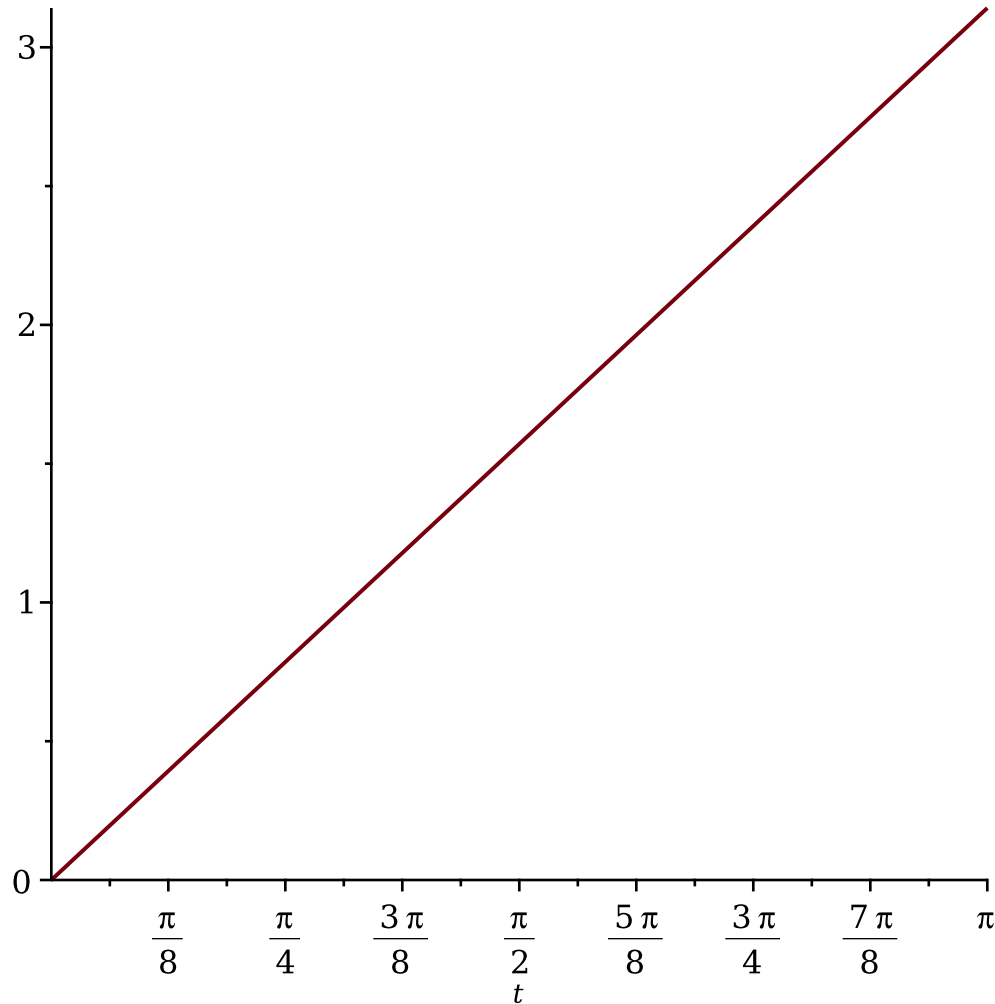
```
> sol:=dsolve({ec,cond},x(t));
```

$$sol := x(t) = t \quad (28)$$

```
> phi:=unapply(rhs(sol),t);
```

$$\phi := t \mapsto t \quad (29)$$

```
> plot(phi(t),t=0..Pi);
```



```
> # b) find solution of the ivp and plot its graph on the interval (Pi, infinity)
```

```
> phid:=t->diff(phi(t),t);
```

$$phid := t \mapsto \frac{d}{dt} \phi(t) \quad (30)$$

```
> ecb:=diff(x(t),t$2)+x(t)=Pi*exp(Pi-t);
```

$$ecb := \frac{d^2}{dt^2} x(t) + x(t) = \pi e^{\pi-t} \quad (31)$$

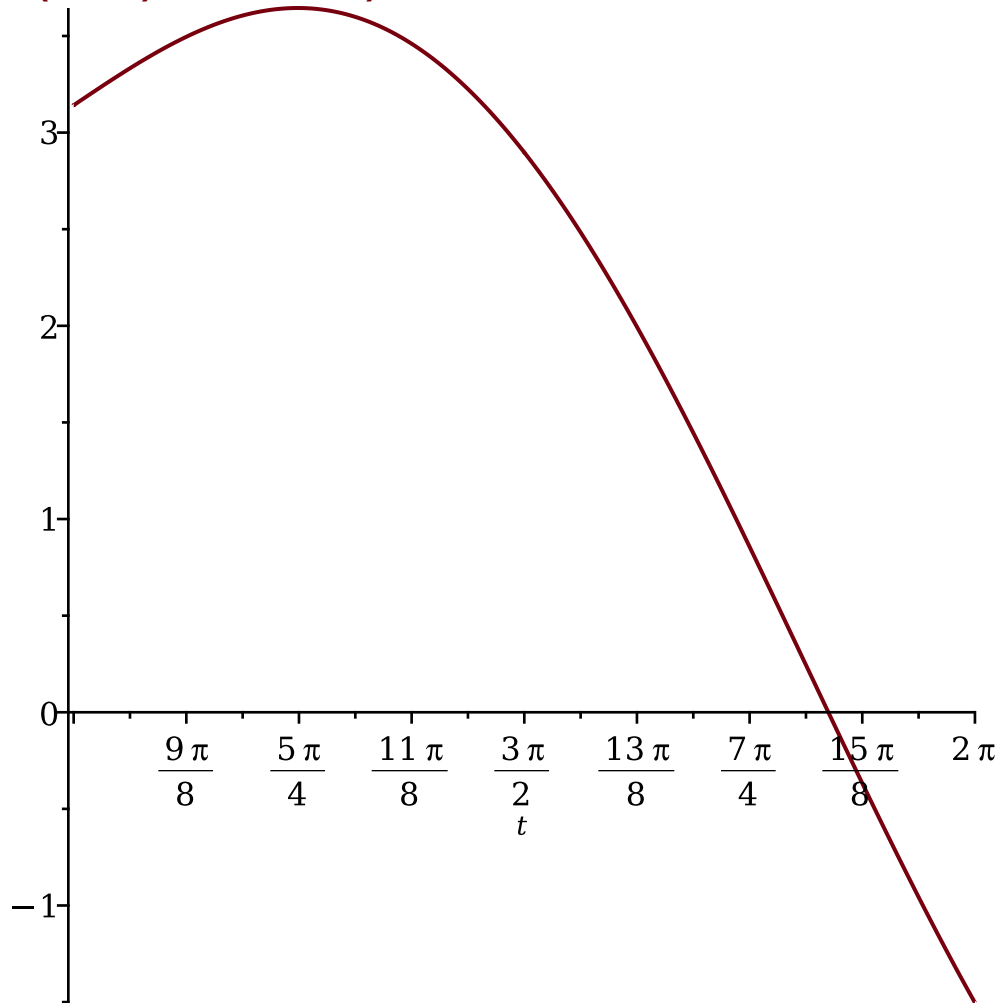
```
> condb:=x(Pi)=phi(Pi),D(x)(Pi)=diff(phi(t),t);
```

$$condb := x(\pi) = \pi, D(x)(\pi) = 1 \quad (32)$$

```
> solb:=dsolve({ecb,condb},x(t));
```

$$\text{solb} := x(t) = \sin(t) \left(-1 - \frac{\pi}{2} \right) - \frac{\cos(t) \pi}{2} + \frac{\pi e^{\pi-t}}{2} \quad (33)$$

```
> plot(rhs(solb),t=Pi..2*Pi);
```



```
> # c) find the solution of the ivp and plot its graph on (0, infinity), f is the function given in ex 3b)
```

```
> f:=t->piecewise(t>=0,t,t>Pi,Pi*exp(Pi-t));
```

$$f := t \mapsto \begin{cases} t & 0 \leq t \\ \pi \cdot e^{\pi-t} & \pi < t \end{cases} \quad (34)$$

```
> ecc:=diff(x(t),t$2)=f(t);
```

$$\text{ecc} := \frac{d^2}{dt^2} x(t) = \begin{cases} t & 0 \leq t \\ \pi e^{\pi-t} & \pi < t \end{cases} \quad (35)$$

```
> condc:=x(0)=0,D(x)(0)=1;
```

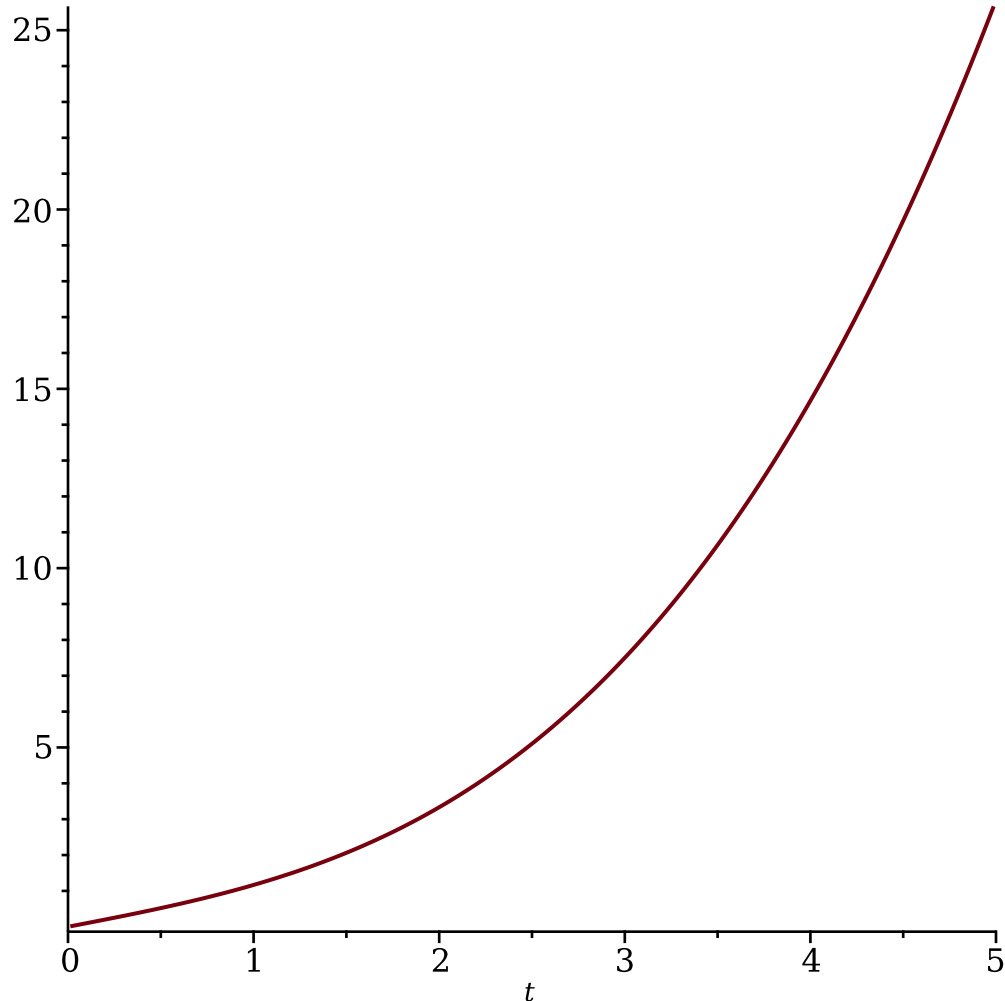
$$\text{condc} := x(0) = 0, D(x)(0) = 1 \quad (36)$$

```
> solc:=dsolve({ecc,condc},x(t));
```

(37)

$$solc := x(t) = \begin{cases} t & t < 0 \\ t + \frac{1}{6} t^3 & 0 \leq t \end{cases} \quad (37)$$

```
> plot(rhs(solc),t=0..5);
```



```
> # d) it glues a,b and c
```

```
> restart;
```

```
> # Exercise 5
```

```
> ec:=diff(x(t),t$2)+x(t)=cos(omega*t);
```

$$ec := \frac{d^2}{dt^2} x(t) + x(t) = \cos(\omega t) \quad (38)$$

```
> cond:=x(0)=0,D(x)(0)=0;
```

$$cond := x(0) = 0, D(x)(0) = 0 \quad (39)$$

```
> # a) dinf fi(t,omega) and note that it's not defined in omega=1
```

```
> sol:=dsolve({ec,cond},x(t));
```

$$sol := x(t) = \frac{\cos(t) - \cos(\omega t)}{\omega^2 - 1} \quad (40)$$

```
> phiomega:=unapply(rhs(sol),t,omega);
```

$$\text{phiomega} := (t, \omega) \mapsto \frac{\cos(t) - \cos(\omega \cdot t)}{\omega^2 - 1} \quad (41)$$

```
> phiomega(t,omega);
```

$$\frac{\cos(t) - \cos(\omega t)}{\omega^2 - 1} \quad (42)$$

```
> # b) find separately the solution when omega=1
```

```
> sol1:=dsolve({diff(x(t),t$2)+x(t)=cos(t),cond},x(t));
```

$$\text{sol1} := x(t) = \frac{\sin(t) t}{2} \quad (43)$$

```
> phires:=unapply(rhs(sol1),t);
```

$$\text{phires} := t \mapsto \frac{\sin(t) \cdot t}{2} \quad (44)$$

```
> phires(t);
```

$$\frac{\sin(t) t}{2} \quad (45)$$

```
> # c) prove the limit for any t
```

```
> lim:=limit(phiomega(t,omega),omega=1);
```

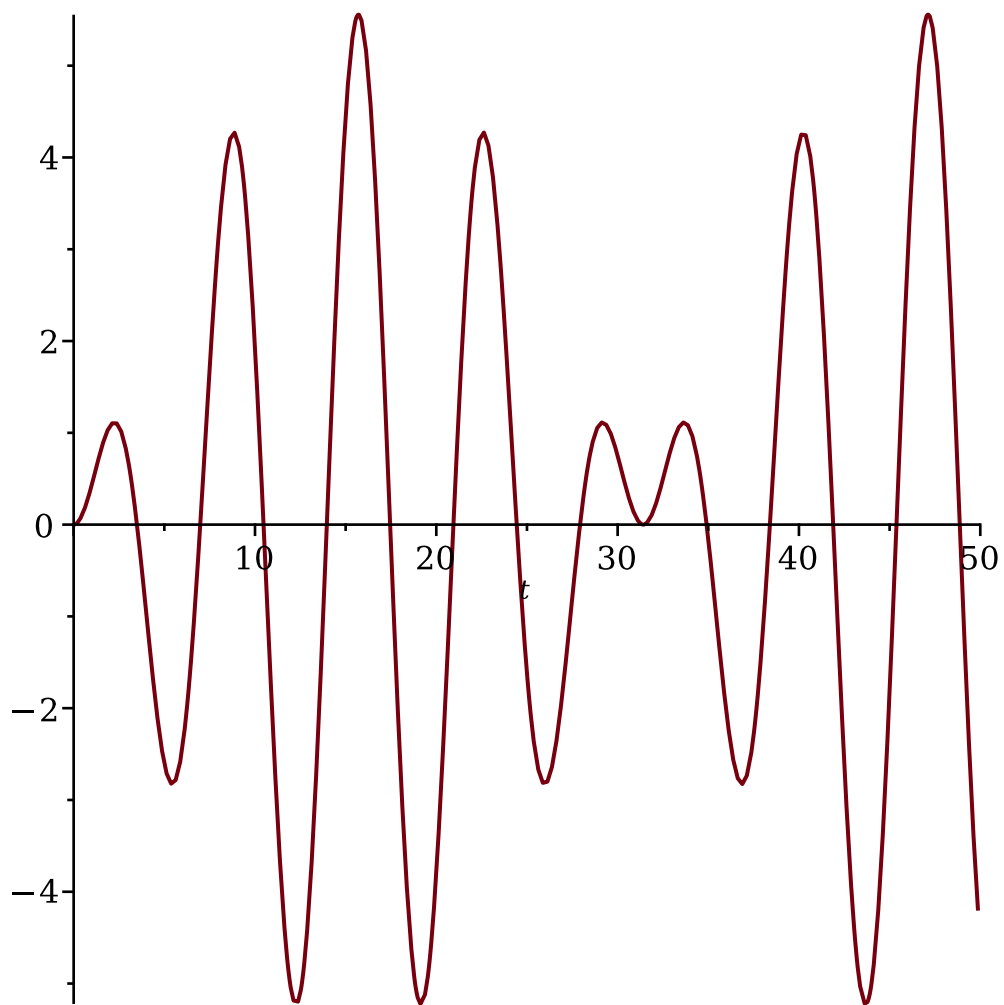
$$\text{lim} := \frac{\sin(t) t}{2} \quad (46)$$

```
> is(lim=phires(t));
```

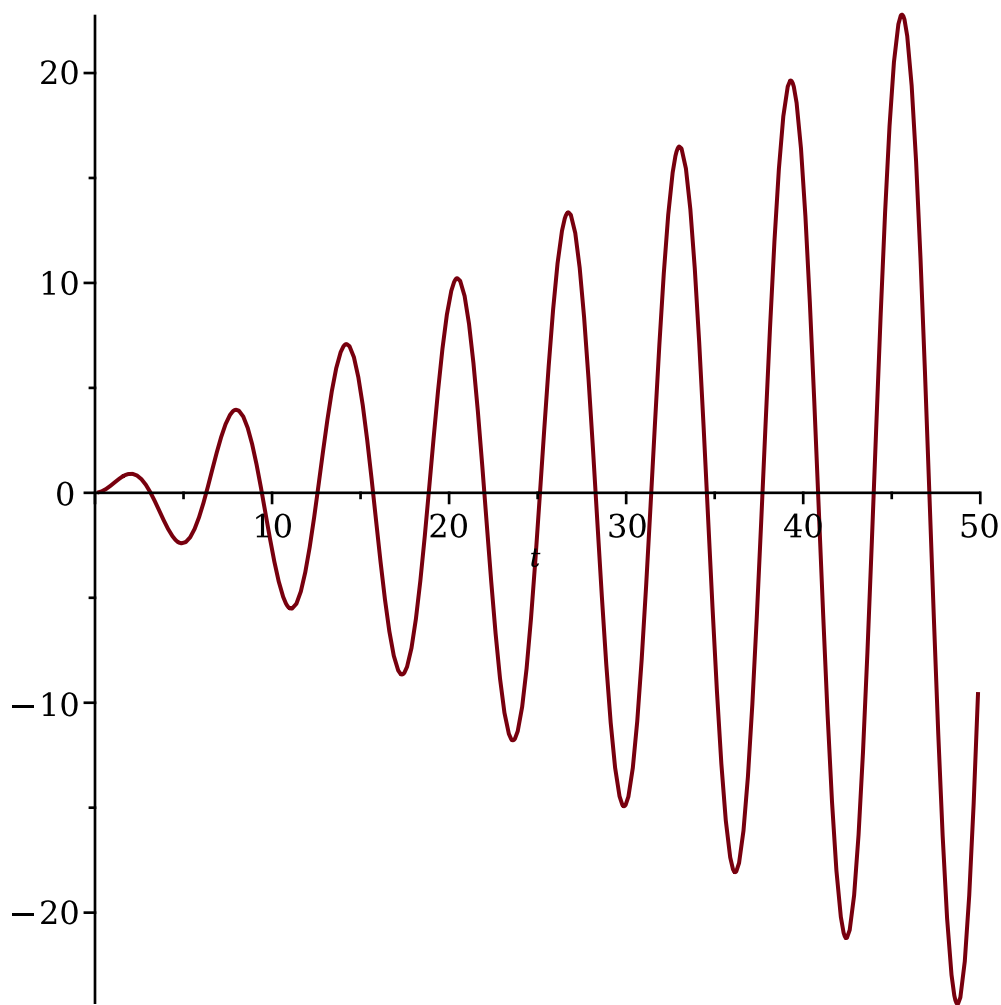
$$\text{true} \quad (47)$$

```
> # d) find bound, plot phi
```

```
> plot(phiomega(t,0.8),t=0..50);
```

```
> plot(phires(t),t=0..50);
```



```
> restart;
```

```
> # Exercise 6
```

```
> ec:=diff(x(t),t$2)-4*x(t)=exp(alpha*t);
```

$$ec := \frac{d^2}{dt^2} x(t) - 4 x(t) = e^{\alpha t} \quad (48)$$

```
> cond:=x(0)=0,D(x)(0)=0;
```

$$cond := x(0) = 0, D(x)(0) = 0 \quad (49)$$

```
> # a) find phi and not that it is not defined in alpha=2
```

```
> sol:=dsolve({ec,cond},x(t));
```

$$sol := x(t) = \frac{(\alpha - 2) e^{-2t} + (-\alpha - 2) e^{2t} + 4 e^{\alpha t}}{4 \alpha^2 - 16} \quad (50)$$

```
> simplify(sol);
```

$$x(t) = \frac{(\alpha - 2) e^{-2t} + (-\alpha - 2) e^{2t} + 4 e^{\alpha t}}{4 \alpha^2 - 16} \quad (51)$$

```
> factor(sol);
```

$$x(t) = - \frac{e^{2t} \alpha - e^{-2t} \alpha + 2 e^{2t} + 2 e^{-2t} - 4 e^{\alpha t}}{4 (\alpha - 2) (\alpha + 2)} \quad (52)$$

```
> # b) solution when alpha = 2
```

```
> ec2:=diff(x(t),t$2)-4*x(t)=exp(2*t);
```

$$ec2 := \frac{d^2}{dt^2} x(t) - 4 x(t) = e^{2t} \quad (53)$$

```
> sol2:=dsolve({ec2,cond},x(t));
```

$$sol2 := x(t) = \frac{(4t-1)e^{2t}}{16} + \frac{e^{-2t}}{16} \quad (54)$$

```
> # c) limit
```

```
> phialpha:=rhs(sol);
```

$$phialpha := \frac{(\alpha-2)e^{-2t} + (-\alpha-2)e^{2t} + 4e^{\alpha t}}{4\alpha^2 - 16} \quad (55)$$

```
> lim:=limit(phialpha,alpha=2);
```

$$lim := -\frac{e^{2t}}{16} + \frac{e^{-2t}}{16} + \frac{te^{2t}}{4} \quad (56)$$

```
> is(lim=rhs(sol2));
```

$$true \quad (57)$$

```
> restart;
```

```
> # Exercise 7
```

```
> f:=t->piecewise(t>=0 and t<Pi/2, t, t>=Pi/2 and t<=Pi, Pi-t, t>Pi
and t<infinity, 0);
```

$$f := t \mapsto \begin{cases} t & 0 \leq t < \frac{\pi}{2} \\ \pi - t & \frac{\pi}{2} \leq t \leq \pi \\ 0 & \pi < t < \infty \end{cases} \quad (58)$$

```
> ec:=diff(x(t),t$2)+x(t)=f(t);
```

$$ec := \frac{d^2}{dt^2} x(t) + x(t) = \begin{cases} t & 0 \leq t < \frac{\pi}{2} \\ \pi - t & \frac{\pi}{2} \leq t \leq \pi \\ 0 & \pi < t < \infty \end{cases} \quad (59)$$

```
> cond:=x(0)=5,D(x)(0)=0;
```

$$cond := x(0) = 5, D(x)(0) = 0 \quad (60)$$

```
> sol:=dsolve({ec,cond},x(t));
```

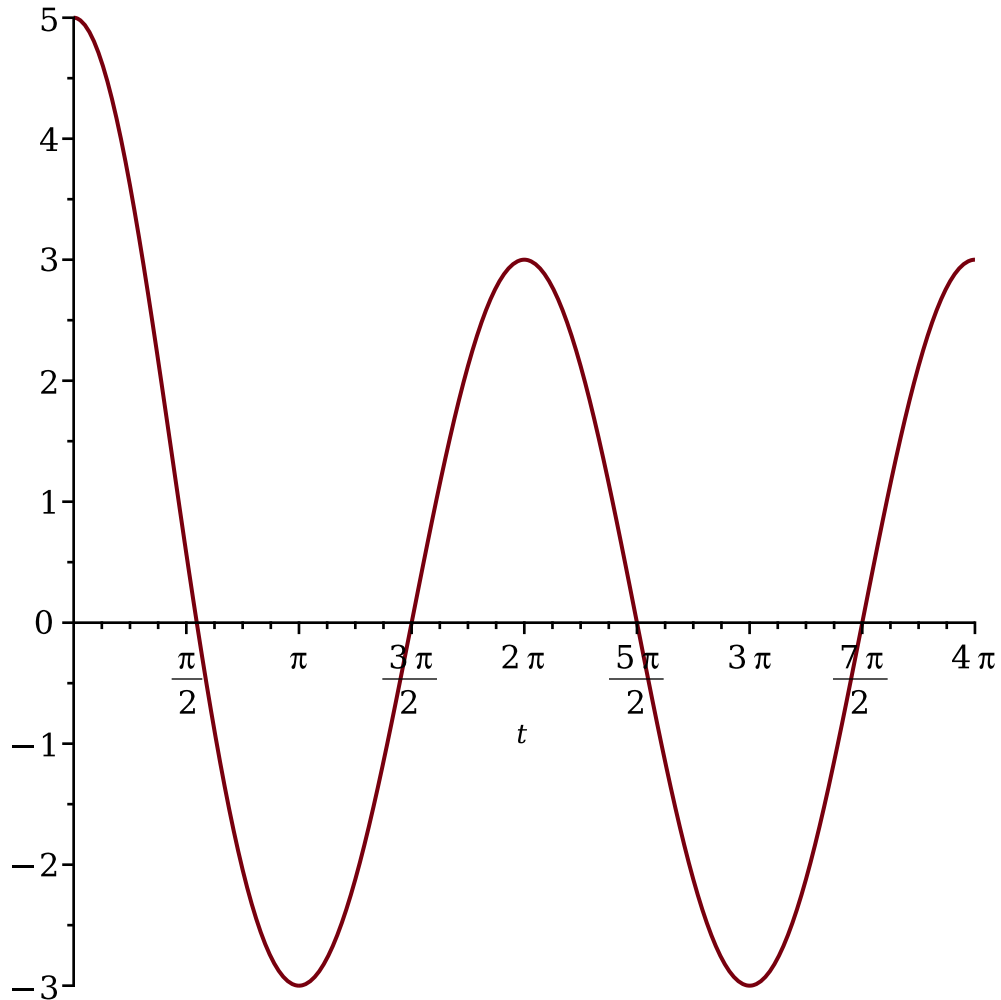
$$(61)$$

$$\text{sol} := x(t) = \begin{cases} 5 \cos(t) & t < 0 \\ 5 \cos(t) + t - \sin(t) & 0 \leq t < \frac{\pi}{2} \\ 3 \cos(t) - \sin(t) + \pi - t & \frac{\pi}{2} \leq t < \pi \\ 3 \cos(t) & \pi \leq t \end{cases} \quad (61)$$

```
> sol;
```

$$x(t) = \begin{cases} 5 \cos(t) & t < 0 \\ 5 \cos(t) + t - \sin(t) & 0 \leq t < \frac{\pi}{2} \\ 3 \cos(t) - \sin(t) + \pi - t & \frac{\pi}{2} \leq t < \pi \\ 3 \cos(t) & \pi \leq t \end{cases} \quad (62)$$

```
> plot(rhs(sol),t=0..4*Pi);
```



```
> restart;
```

```
> # Exercise 8
```

```
> with(plots);
```

[*animate, animate3d, animatecurve, arrow, changecoords, complexplot, complexplot3d, conformal, conformal3d, contourplot, contourplot3d, coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot, fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal, interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d, listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot, multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot, polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot, rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween, spacecurve, sparsematrixplot, surfdata, textplot, textplot3d, tubeplot*]

(63)

> **sys:={diff(x(t),t)=-2*x(t),diff(y(t),t)=-3*y(t)};**

$$\text{sys} := \left\{ \frac{d}{dt} x(t) = -2x(t), \frac{d}{dt} y(t) = -3y(t) \right\}$$

(64)

> **ic:={x(0)=1,y(0)=1};**

$$ic := \{x(0) = 1, y(0) = 1\}$$

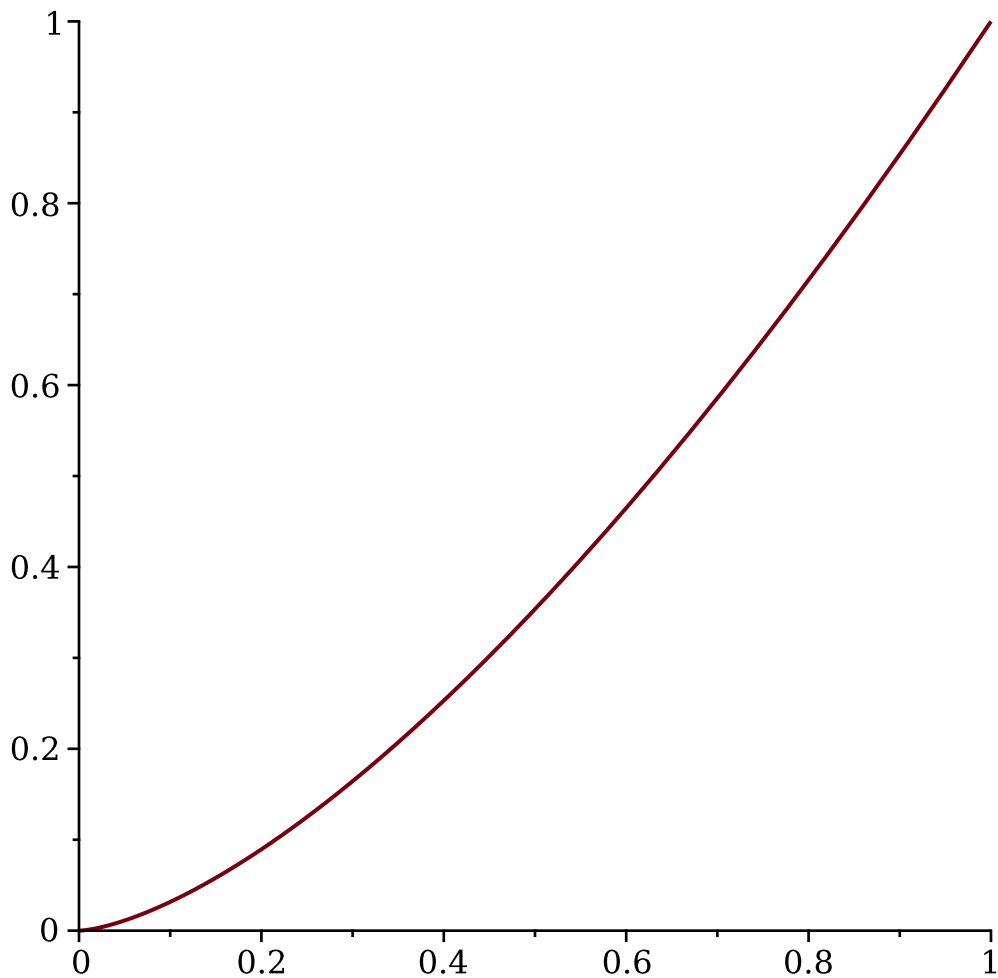
(65)

> **sol:=dsolve(sys union ic);**

$$sol := \{x(t) = e^{-2t}, y(t) = e^{-3t}\}$$

(66)

> **plot([rhs(sol[1]),rhs(sol[2]),t=0..5]);**



```
> restart;
```

```
> # b)
```

```
> with(plots);
```

```
[animate, animate3d, animatecurve, arrow, changecoords, complexplot,
complexplot3d, conformal, conformal3d, contourplot, contourplot3d,
coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot,
fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal,
interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d,
listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot,
multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot,
polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot,
rootlocus, semilogplot, setcolors, setoptions, setoptions3d,
shadebetween, spacecurve, sparsematrixplot, surfdata, textplot,
textplot3d, tubeplot]
```

(67)

```
> sys:={diff(x(t),t)=-2*x(t),diff(y(t),t)=3*y(t)};
```

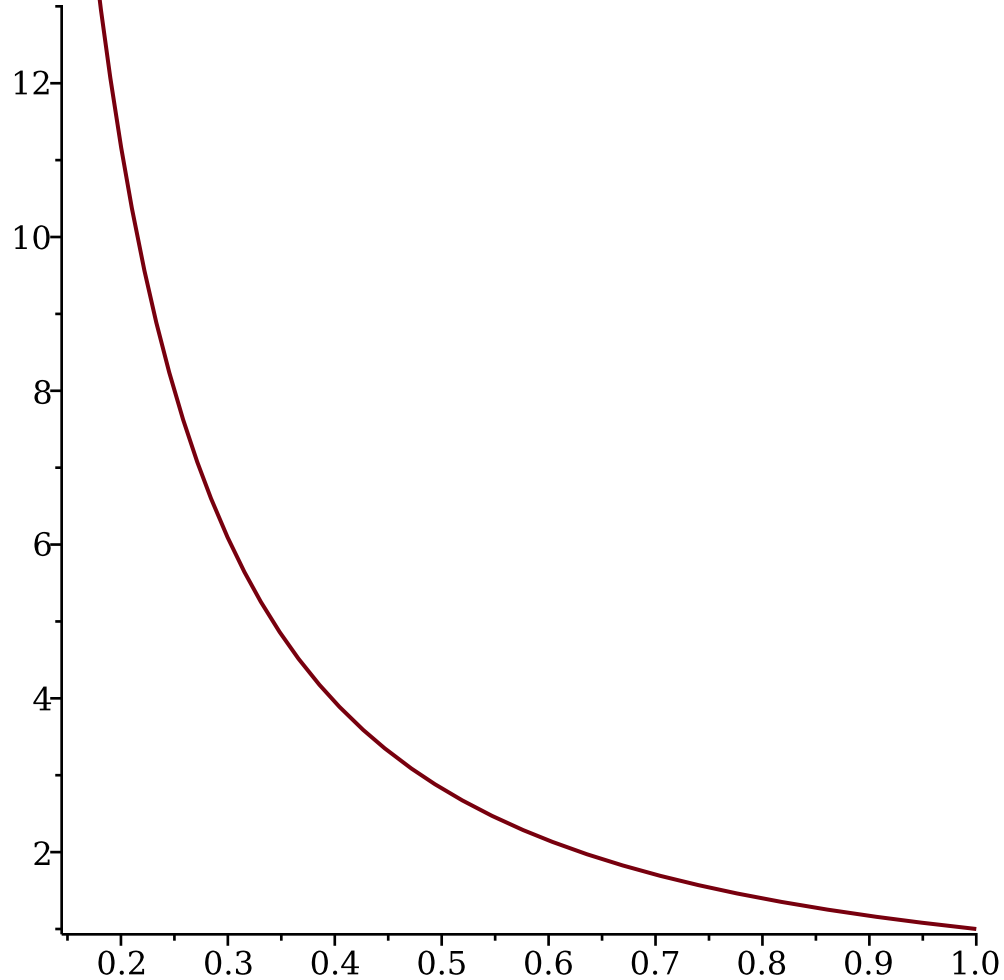
$$\text{sys} := \left\{ \frac{d}{dt} x(t) = -2 x(t), \frac{d}{dt} y(t) = 3 y(t) \right\}$$

(68)

```
> ic:={x(0)=1,y(0)=1};
                                 $ic := \{x(0) = 1, y(0) = 1\}$  (69)
```

```
> sol:=dsolve(sys union ic);
                                 $sol := \{x(t) = e^{-2t}, y(t) = e^{3t}\}$  (70)
```

```
> plot([rhs(sol[1]),rhs(sol[2]),t=0..5]);
```



```
> restart;
```

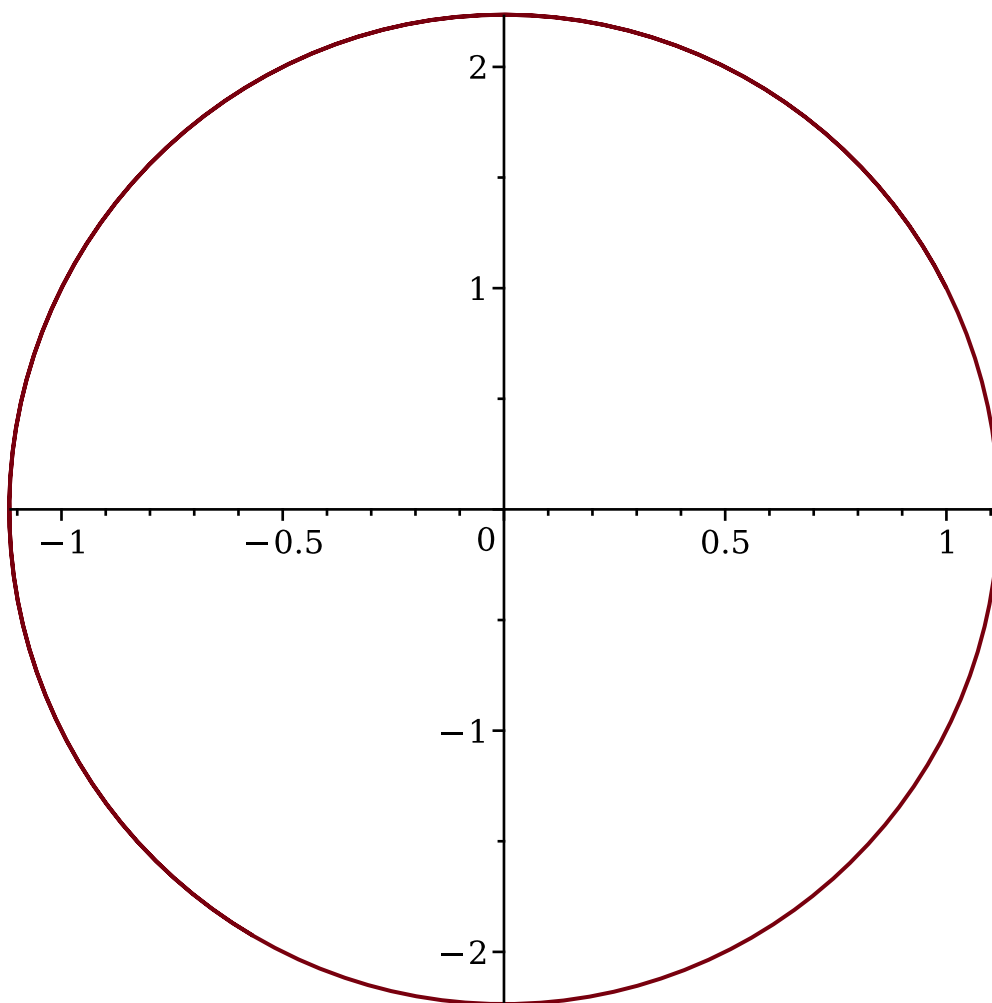
```
> # c)
```

```
> sys:={diff(x(t),t)=-y(t),diff(y(t),t)=4*x(t)};
                                 $sys := \left\{ \frac{d}{dt} x(t) = -y(t), \frac{d}{dt} y(t) = 4x(t) \right\}$  (71)
```

```
> ic:={x(0)=1,y(0)=1};
                                 $ic := \{x(0) = 1, y(0) = 1\}$  (72)
```

```
> sol:=dsolve(sys union ic);
                                 $sol := \left\{ x(t) = -\frac{\sin(2t)}{2} + \cos(2t), y(t) = \cos(2t) + 2\sin(2t) \right\}$  (73)
```

```
> plot([rhs(sol[1]),rhs(sol[2]),t=0..5]);
```



```

> restart;
> # d)
> sys:={diff(x(t),t)=-x(t)+3*y(t),diff(y(t),t)=-3*x(t)-y(t)};
      sys := {  $\frac{d}{dt} x(t) = -x(t) + 3y(t), \frac{d}{dt} y(t) = -3x(t) - y(t)$  }

```

(74)

```

> ic:={x(0)=1,y(0)=1};
      ic := {x(0) = 1, y(0) = 1}

```

(75)

```

> sol:=dsolve(sys union ic);
      sol := {x(t) = e-t (cos(3t) + sin(3t)), y(t) = e-t (cos(3t) - sin(3t))}

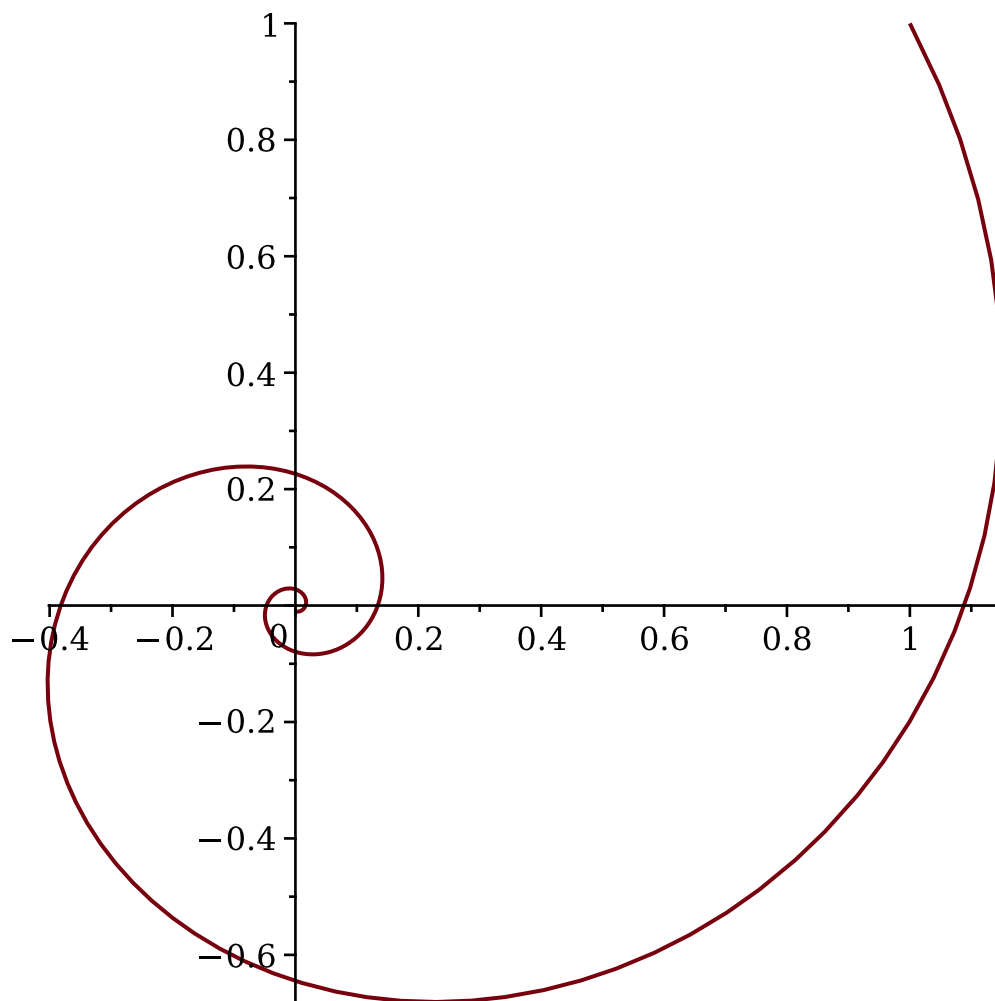
```

(76)

```

> plot([rhs(sol[1]),rhs(sol[2]),t=0..5]);

```

```

> restart;
> # d)
> sys:={diff(x(t),t)=-x(t)+3*y(t),diff(y(t),t)=3*x(t)-y(t)};
      sys := {  $\frac{d}{dt} x(t) = -x(t) + 3y(t), \frac{d}{dt} y(t) = 3x(t) - y(t)$  }

```

(77)

```

> ic:={x(0)=1,y(0)=1};
      ic := {x(0) = 1, y(0) = 1}

```

(78)

```

> sol:=dsolve(sys union ic);
      sol := {x(t) = e2t, y(t) = e2t}

```

(79)

```

> plot([rhs(sol[1]),rhs(sol[2]),t=0..5]);

```

